FastAPI Routing & Architecture Handbook

FastAPI Code Routing & Modular Architecture

A Complete Technical Masterclass on APIRouter, Path Parameters, Dependency Injection, Middleware Routing, and Modular Microservices Code Design

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Table of Contents

Module 1: Fundamentals of Web Routing & HTTP Methods in FastAPI

Module 2: Path Parameters, Query Parameters & Body Payloads

Module 3: Modular Application Design with APIRouter

Module 4: Project Directory Structure & Clean Routing Architecture

Module 5: Route Dependencies & Dependency Injection Hierarchy

Module 6: Dynamic Path Routing, Enum Matching & Regex Parameters

Module 7: Request Validation, Status Codes & Custom Responses

Module 8: Error Handling, Global Handlers & Middleware Routing

Module 9: WebSockets, Streaming & Asynchronous Background Task Routing

Module 10: Production Routing Strategies, API Versioning & OpenAPI Tags

Module 1: Fundamentals of Web Routing & HTTP Methods

1.1 What is Routing in Web Frameworks?

Routing is the mechanism of binding HTTP endpoints (URIs and HTTP Verbs) to Python execution functions (Route Handlers or Path Operation Functions). FastAPI routes requests efficiently using Starlette ASGI routing tree.

1.2 Basic Route Definition Example

```
from fastapi import FastAPI

app = FastAPI(title="Core Routing Engine")

@app.get("/")
async def root():
    return {"message": "Welcome to Core FastAPI Routing Engine"}

@app.post("/items")
async def create_item(payload: dict):
    return {"status": "created", "data": payload}
```

Module 2: Path Parameters, Query Parameters & Body Payloads

2.1 Understanding Path vs Query Parameters

Path parameters are required dynamic segment variables within the URL path. Query parameters are key-value pairs following the URL question mark (`?`).

2.2 Comprehensive Parameter Routing Example

```
from fastapi import FastAPI, Path, Query, Body
from typing import Optional

app = FastAPI()

@app.get("/users/{user_id}/posts/{post_id}")
async def get_user_post(
    user_id: int = Path(..., title="The ID of the user", ge=1),
    post_id: str = Path(..., title="The post slug identifier"),
    search: Optional[str] = Query(None, max_length=50),
    page: int = Query(1, ge=1)
):
    return {
        "user_id": user_id,
        "post_id": post_id,
        "search_query": search,
        "page": page
    }
```

Module 3: Modular Application Design with APIRouter

3.1 Why Use APIRouter?

In large-scale applications, defining all routes inside a single `main.py` file leads to monolithic codebase sprawl. `APIRouter` allows splitting endpoints into domain-driven modular files.

3.2 Modular Router Setup

```
# app/api/v1/endpoints/users.py
from fastapi import APIRouter

router = APIRouter(prefix="/users", tags=["User Management"])

@router.get("/")
async def list_users():
    return [{"user_id": 1, "name": "Alice"}, {"user_id": 2, "name": "Bob"}]

@router.get("/{user_id}")
async def get_user(user_id: int):
    return {"user_id": user_id, "name": "Alice"}
```

Module 4: Project Directory Structure & Clean Routing

4.1 Production Folder Structure

```
my_fastapi_app/
├── app/
│   ├── __init__.py
│   ├── main.py
│   ├── api/
│   │   ├── __init__.py
│   │   ├── router.py
│   │   └── v1/
│   │       ├── __init__.py
│   │       ├── auth.py
│   │       ├── users.py
│   │       └── products.py
│   ├── core/
│   │   └── config.py
│   └── services/
└── requirements.txt
```

4.2 Combining Routers in Main Entrypoint

```
# app/main.py
from fastapi import FastAPI
from app.api.v1.users import router as user_router
from app.api.v1.auth import router as auth_router

app = FastAPI(title="Enterprise API Service")

# Registering Routers
app.include_router(auth_router, prefix="/api/v1/auth", tags=["Auth"])
app.include_router(user_router, prefix="/api/v1/users", tags=["Users"])
```

Module 5: Route Dependencies & Dependency Injection Hierarchy

5.1 Multi-Level Dependency Injection

Dependencies can be injected at the global application level, router level, or individual endpoint function level.

```
from fastapi import APIRouter, Depends, Header, HTTPException

async def verify_token(x_api_key: str = Header(...)):
    if x_api_key != "secret-token":
        raise HTTPException(status_code=400, detail="Invalid API Key Header")

router = APIRouter(
    prefix="/admin",
    tags=["Admin"],
    dependencies=[Depends(verify_token)]
)

@router.get("/dashboard")
async def admin_dashboard():
    return {"status": "Welcome to protected admin panel"}
```

Module 6: Dynamic Path Routing, Enum Matching & Regex

6.1 Enum Constrained Path Operations

```
from enum import Enum
from fastapi import FastAPI

class ModelName(str, Enum):
    alexnet = "alexnet"
    resnet = "resnet"
    lenet = "lenet"

app = FastAPI()

@app.get("/models/{model_name}")
async def get_model(model_name: ModelName):
    if model_name is ModelName.alexnet:
        return {"model_name": model_name, "message": "Deep Learning Classic!"}
    return {"model_name": model_name, "message": "Other ML Architecture"}
```

Module 7: Request Validation, Status Codes & Custom Responses

7.1 Response Types & Custom Status Codes

```
from fastapi import FastAPI, status
from fastapi.responses import JSONResponse, RedirectResponse

app = FastAPI()

@app.post("/items", status_code=status.HTTP_201_CREATED)
async def create_item(name: str):
    return {"message": "Item created successfully", "item_name": name}

@app.get("/old-path")
async def redirect_old_route():
    return RedirectResponse(url="/new-path", status_code=status.HTTP_301_MOVED_PERMANENTLY)
```

Module 8: Error Handling, Global Handlers & Middleware Routing

8.1 Custom Exception Router Intercepts

```
from fastapi import FastAPI, Request, HTTPException
from fastapi.responses import JSONResponse

app = FastAPI()

class ItemNotFoundException(Exception):
    def __init__(self, name: str):
        self.name = name

@app.exception_handler(ItemNotFoundException)
async def custom_item_not_found_handler(request: Request, exc: ItemNotFoundException):
    return JSONResponse(
        status_code=404,
        content={"message": f"Oops! The item '{exc.name}' does not exist in inventory."}
    )
```

Module 9: WebSockets, Streaming & Asynchronous Task Routing

9.1 WebSocket Routing

```
from fastapi import FastAPI, WebSocket, WebSocketDisconnect

app = FastAPI()

@app.websocket("/ws/chat/{client_id}")
async def websocket_endpoint(websocket: WebSocket, client_id: str):
    await websocket.accept()
    try:
        while True:
            data = await websocket.receive_text()
            await websocket.send_text(f"Client #{client_id} says: {data}")
    except WebSocketDisconnect:
        print(f"Client #{client_id} disconnected")
```

Module 10: Production Routing Strategies & API Versioning

10.1 Multi-Version Routing Strategy

```
from fastapi import FastAPI, APIRouter

app = FastAPI(title="Versioned Enterprise API")

v1_router = APIRouter(prefix="/v1")
v2_router = APIRouter(prefix="/v2")

@v1_router.get("/users")
async def get_users_v1():
    return [{"id": 1, "name": "John Doe"}]

@v2_router.get("/users")
async def get_users_v2():
    return [{"id": 1, "first_name": "John", "last_name": "Doe", "version": "v2"}]

app.include_router(v1_router)
app.include_router(v2_router)
```